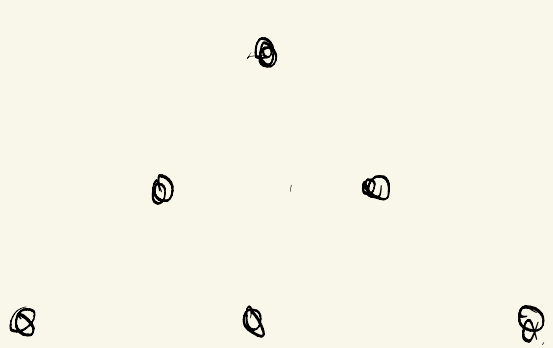
Eco 202 – Fall 2026

Week 10



$$X_j \in \{0,1\} \implies \mathbb{P}(X_j = 1) = \mathbb{P}(X_j = 0) = \frac{1}{2}$$

$$\mathbb{P}(X_1 = x_1, X_2 = x_2, \ldots, X_J = x_J)$$

$X_j \sim$ Bernoulli $(p = 1/2)$

$$Y = \sum_{j=1}^{J} X_j \sim \text{Binomial}(p = 1/2, J)$$

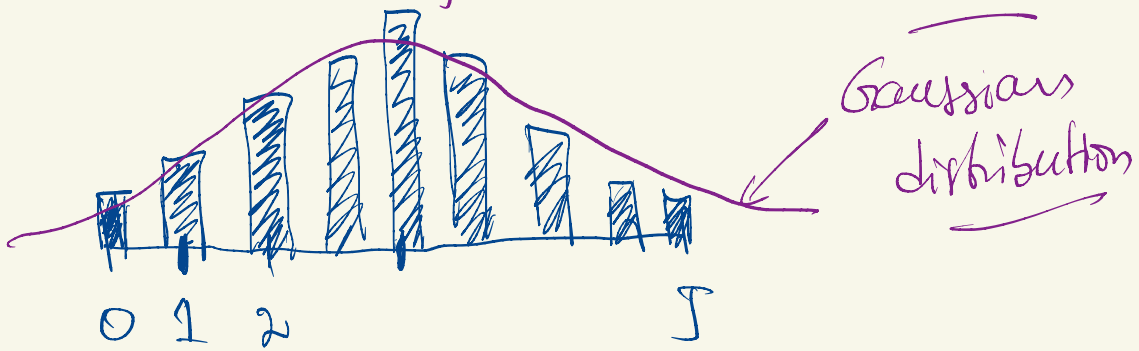
J: # of trials

$$X_1 \sim \text{Bernoulli}(p=1/2)$$
$$X_2 \sim \text{Bernoulli}(p=1/2)$$
$$\vdots$$
$$X_J \sim \text{Bernoulli}(p=1/2)$$

} independent

$$Y = \sum_{j=1}^{J} X_j \sim \text{Binomial}(p=1/2, J)$$

$X_j \in \{0,1\}$ (support) $\Rightarrow$ $Y \in \{0, 1, 2, \ldots, J\}$



- 2 random variables : X, Y, $x \in \mathcal{X}$

  - $\mathbb{E}[X] = \sum_{x \in \mathcal{X}} x \cdot \mathbb{P}[X = x]$

Example : $\mathcal{X} = \{0, 1\} \Rightarrow X \sim \text{Bernoulli}(\mathbb{P}(X=1))$

$$\mathbb{E}[X] = 0 \cdot \mathbb{P}[X=0] + 1 \cdot \mathbb{P}[X=1] = \mathbb{P}(X=1)$$

- $\mathbb{V}[X] = \mathbb{E}\left[(X - \mathbb{E}[X])^2 \right]$

$$= \sum_{x \in \mathcal{X}} (x - \mathbb{E}[X])^2 \cdot \mathbb{P}(X = x)$$

Example : $\mathcal{X} = \{0, 1\} \Rightarrow X \sim \text{Bernoulli}(\mathbb{P}(X=1))$

$$\mathbb{V}[X] = (0 - \mathbb{P}(X=1))^2 \cdot \mathbb{P}(X=0) + (1 - \mathbb{P}(X=1))^2 \cdot \mathbb{P}(X=1)$$

$$= \mathbb{P}(X=1)(1 - \mathbb{P}(X=1)) \quad \text{(check)}$$

- Recall: 2 random variables X, Y

$$\mathbb{E}[X+Y] = \mathbb{E}[X] + \mathbb{E}[Y]$$

$$\mathbb{V}[X+Y] = \mathbb{V}[X] + \mathbb{V}[Y] + 2\,\mathrm{Cov}(X,Y)$$

If X and Y are <u>independent</u> then $\mathrm{Cov}(X,Y) = 0$.

<u>Recall</u>:

$$\mathrm{Cov}(X,Y) = \mathbb{E}\left[(X-\mathbb{E}[X])(Y-\mathbb{E}[Y])\right]$$

<u>Note</u>: $\mathrm{Cov}(X,X) = \mathbb{V}[X]$.